

STACKLY / PROJECT DOCUMENTATION

ROBOTICS & AUTOMATION

System Specification • UI/UX • Architecture
Dashboards • Animation • Performance

DOCUMENT

STACKLY Robotics & Automation

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27 source pages

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STACKLY Robotics & Automation

Introduction & System Overview

This project is a premium robotics and automation platform built to deliver a bold, high-contrast digital experience that empowers manufacturers, plant engineers, and operators to explore autonomous work cells, review deployment capabilities, and manage live fleet operations with confidence. The system seamlessly connects a polished, multi-page public-facing website with powerful internal consoles, enabling visitors to explore systems, review real-world deployments, read technical guides, request a consult, and manage fleet telemetry — all while maintaining an industrial, control-room brand aesthetic that reads as precise, reliable, and engineered.

Behind the scenes, the architecture integrates a clean, semantic HTML5 layout with advanced CSS styling — including responsive grid systems, CSS custom properties, and modern layout techniques — alongside robust JavaScript functionality such as dynamic form validation, custom dropdown components, role-based routing with `localStorage` session management, live telemetry widgets, interactive ROI/case-study counters, article category filtering, Chart.js-powered dashboards, an infinite category filter, a full-screen mobile drawer, and a 404 signal-lost page. Two signature GSAP animation systems — **Gravity Fall** (characters drop in from above with a `bounce.out` easing) and **Popcorn Pop** (headings pop into place character by character with a random rotation) — combined with the **Lenis smooth-scroll** engine keep the browsing experience feeling mechanical and alive across every screen size.

Technology Stack & Dependencies

Core Technologies

HTML5 & CSS3: The foundational building blocks used to structure the semantic page content and style the modern, responsive layout. Utilizes CSS custom properties (variables) to maintain a consistent design system featuring the

signature Signal Yellow (#DBFE01), White (#FFFFFF), Pure Black (#000000) and supporting charcoal/surface palette, alongside advanced techniques like CSS Grid for the console layouts, editorial service grids, and Flexbox for responsive content organization. The design uses layered gradients, neon-accented borders, and industrial blueprint compositions to maintain a premium autonomous-technology aesthetic.

Vanilla JavaScript: Lightweight, native scripting utilized to handle real-time user inputs, complex form validation (login, signup, contact with custom validation), click events, tab-driven view switching in dashboards, custom dropdown components, password reveal toggles, interactive FAQ accordions, `localStorage` session management for demonstration authentication, role-based portal routing, article category filtering, interactive ROI counters, GSAP letter-splitting for hero titles and section headings, and a full-screen mobile drawer — without the need for heavy third-party frameworks.

Lenis Smooth Scroll: A lightweight smooth-scroll engine loaded via CDN and initialised in a shared `lenis-init.js` file. Lenis replaces the browser's native scroll physics with a lerped `requestAnimationFrame` loop using a custom easing curve, giving the entire platform its signature glide. All pages that use Lenis tag `html.lenis` and manage `lenis-stopped` state during menu/drawer opening to prevent scroll bleed.

External Libraries & CDNs

GSAP (v3.12.5+) & ScrollTrigger: Loaded externally via CDN to power complex, scroll-driven motion sequences, including the **Gravity Fall** hero text animation (characters dropping in from `y: -300` with `bounce.out`), the **Popcorn Pop** section heading animation (character-by-character scale-up with random rotation and `back.out` easing), the entrance timelines on auth pages, and viewport-triggered reveals across the site.

Font Awesome Icons (v6.5.1): Included via CDN to provide a vast library of scalable vector icons used consistently across navigation, buttons, social links, service cards, dashboard statuses, work order rows, and interactive elements throughout the platform.

Material Symbols Outlined: Included in the operator console to provide a modern, variable-weight icon system for the sidebar navigation, toolbars, and quick-action

cards.

Chart.js: Included in the operator console via CDN to render the site activity line chart and the work-order status donut chart on the Control Room dashboard.

Google Fonts (Outfit + Zen Maru Gothic + Retro Floral): Uses Outfit for clean geometric body copy, Zen Maru Gothic for the navigation and badges, and Retro Floral (with an Outfit fallback) for the large display headings conveying industrial weight.

File Architecture & Directory Structure

The project is structured as a Multi-Page Application (MPA), where each separate HTML file serves as an independent, dedicated screen or page view. All pages are organized within a clean, logical directory structure that separates assets, styles, scripts, and HTML files for maintainability and scalability.

```

stackly-robotics/
├── index.html                # Root entry point (Home)
├── 404.html                 # Custom error page (Signal Lost)
├── global.css               # Global design tokens & shared components
├── index.css                # Home page & shared page styles
├── lenis-init.js            # Shared Lenis smooth-scroll initializer
├── logo.webp                # Brand logo asset
├── assets/                  # WebP photography (hero, machines, robots, team)
│   ├── home/               # Home page photography (hero, machine, arm, transfor
m, b_*)
│   ├── about/              # About page photography (hero, how_it_works, what_we
_*, sphere, facility, leadership, team-*)
│   ├── service/            # Services page photography (hero, smart, floor, end)
│   ├── contact/            # Contact page photography (left_top, left_bottom, ce
nter, right_top, right_bottom)
│   ├── blog/               # Blog page photography (hero, latest_*, how, blog_ct
a_bg)
│   ├── remote/             # Remote shared photography (remote_*)
│   ├── login_page.webp
│   └── signup_page.webp
├── css/                     # Page-specific stylesheets
│   ├── about.css           # About page styles
│   ├── blog.css            # Blog page styles
│   ├── contact.css         # Contact page styles
│   ├── login.css           # Login page styles (also base for signup)
│   ├── services.css        # Services page styles
│   └── signup.css          # Signup page styles
└── html/                    # All HTML pages (excluding root index)
    └── about.html          # About page

```

```

├─ admin.html          # Fleet Command dashboard (Admin console)
├─ blog.html           # Blog page
├─ contact.html        # Contact page
├─ login.html          # Login page
├─ services.html       # Services page
├─ signup.html         # Signup page
└─ user.html           # Operator Console dashboard (User)

```

Directory Breakdown

Folder	Purpose	Key Files
Root	Core entry points and global styles	index.html, global.css, index.css, lenis-init.js, logo.webp, 404.html
/assets/	Optimized WebP photography	home/, about/, service/, contact/, blog/, remote/, login_page.webp, signup_page.webp
/css/	Page-specific stylesheets	about.css, blog.css, contact.css, login.css, services.css, signup.css
/html/	All HTML pages (excluding root index)	about.html, admin.html, blog.html, contact.html, login.html, services.html, signup.html, user.html

Visual Design System (UI/UX)

Modern CSS Grid & Flexbox

Implements responsive layouts and complex grid structures via `display: grid` and `display: flex` to seamlessly organize content across all screen sizes. The platform uses custom breakpoints at 1200px, 1100px, 1024px, 992px, 900px, 820px, 768px, 680px, 640px, 600px, 560px, 550px, 480px, 380px and 300px for optimal device adaptation. Editorial service grids, industry layouts, dashboard KPI rows, chart panels, cell-solution grids, timeline spheres, and driver/operator card grids all leverage these modern layout techniques to maintain visual harmony across devices.

Directional Slide Reveal System

Unlike conventional fade-in reveals, Stackly uses a four-directional slide reveal system driven by a `[data-anim]` attribute:

- `[data-anim="left"]` — `translate3d(-56px, 0, 0)`
- `[data-anim="right"]` — `translate3d(56px, 0, 0)`
- `[data-anim="up"]` — `translate3d(0, 44px, 0)`
- `[data-anim="down"]` — `translate3d(0, -44px, 0)`
- `[data-anim="fade"]` — no transform

Elements are activated by an `IntersectionObserver` that adds the `is-visible` class, releasing the transform and easing opacity back to 1. The transition uses `cubic-bezier(0.22, 1, 0.36, 1)` over 0.9s. Per-element delays are applied via a `data-delay` attribute (in seconds).

Gravity Fall Hero Animation

The hero title characters on the home page and about page are split into individual `<span class="hero-char">` elements by JavaScript and animated with a GSAP `from()` using `y: -300, opacity: 0, stagger: 0.04, duration: 0.8, ease: 'bounce.out'`. This produces a distinctive mechanical typing effect as if each letter falls into place and rebounds — reinforcing the industrial, robotic tone of the brand.

Popcorn Pop Section Heading Animation

Section headings across the site are split into individual character `<span class="popcorn-char">` elements and animated in with a GSAP `from()` using `scale: 0, y: 30, rotation: random(-20, 20), stagger: { each: 0.04, from: 'random' }, duration: 0.4, ease: 'back.out(2)'`. This creates a distinctive "popcorn" effect where characters jump into place, reinforcing the operational, energy-driven tone of the brand.

Interactive Micro-Animations

Built with smooth CSS transitions using a single shared easing curve (`cubic-bezier(0.22, 1, 0.36, 1)` or `cubic-bezier(0.2, 0.8, 0.2, 1)` depending on context) and subtle transforms (`transform: translateY(-6px)` on cards, `transform: scale(1.05)` on image hover, `padding-left` shifts on editorial rows, `transform: rotate(90deg)` on close buttons) to give immediate visual feedback on buttons, cards, image figures, industry list items, FAQ summaries, and dashboard controls. These restrained interactions create a precise, professional user experience that feels intentional and industrial.

Full-Screen Mobile Drawer

The public site uses a full-screen slide-in mobile drawer (`right: -100vw` → `right: 0`) with a blurred dark backdrop, body scroll locking (`body.menu-open`), Escape key dismissal, click-outside dismissal, and an automatic close when the viewport grows beyond 900px. Both admin and operator dashboards use a similar pattern: fixed sidebar slides off-screen (`translateX(-100%)`) on mobile, opening via a hamburger toggle with a scrim overlay.

Bold Industrial Aesthetic

The visual design prioritizes speed, reliability, and operational confidence through:

- **Bold palette:** Signal Yellow (#DBFE01) for primary CTAs, accents and highlights; White (#FFFFFF) for text on dark surfaces; Pure Black (#000000) for the primary page background; Deep Surface (#0A0A0A) for elevated panels; Card (#111111 / #181818 / #1E1E1E) for dashboard cards
- **Industrial typography:** Retro Floral (with Outfit fallback) 900 weight for display headings with tight letter-spacing, Zen Maru Gothic for nav and badges, Outfit for clean body copy
- **Neon glow effects:** Yellow text-shadows on hero titles, box-shadows on buttons, drop-shadows on robotic arm imagery
- **Layered gradient scrim:** Hero overlays, dashboard headers, CTA banners and glassmorphism panels all use carefully-tuned linear and radial gradients
- **Full-screen dramatic elements:** Neon yellow wedges with clip-path, floating wireframe spheres, giant blueprint typography, and 5-ring contact hero stages
- **Photography-led layouts:** Industrial, warehouse, and robotic arm imagery allowed to dominate hero and services sections

Visual Design Language & Color Palette

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
Signal Yellow	--primary-yellow	#DBFE01	Primary CTAs, accents, eyebrows, active states, chart series

Color Name	CSS Variable	Hex / RGBA Code	Usage Context
White	--white	#FFFFFF	Body text on dark surfaces, headings
Black	--bg-black	#000000	Primary page background
Surface 1	--surface-1	#0A0A0A	Elevated panels, sections
Surface 2	--surface-2	#111111	Dashboard cards, sidebar
Card	--color-card	#1e1f24	Dashboard card background
Line Subtle	--line-subtle	rgba(255, 255, 255, 0.10)	Borders on dark surfaces
Line Strong	--line-strong	rgba(255, 255, 255, 0.20)	Stronger borders, form input borders
Text Secondary	--text-secondary	rgba(255, 255, 255, 0.68)	Secondary body copy
Text Muted	--text-muted	rgba(255, 255, 255, 0.42)	Metadata, labels, eyebrow text
Error State	—	#ff6b5b / #FF4D4D / #ff5247	Invalid form fields, error text
Success State	--status-online	#22c55e	Positive status indicators
Warning State	--status-standby	#f59e0b	Warning status indicators
Alert State	--status-alert	#ef4444	Error status indicators
Info State	--status-info	#38bdf8	Info status indicators
AI State	--status-ai	#a855f7	AI-specific status indicators

Typography System

Font Name	CSS Variable	Font Family	Usage Context
Display Font	--font-retro	"Retro Floral", "Outfit", sans-serif	Display headings, hero titles, section titles, editorial statements
Nav Font	--font-nav	"Zen Maru Gothic", sans-serif	Navigation, badges, buttons, meta labels

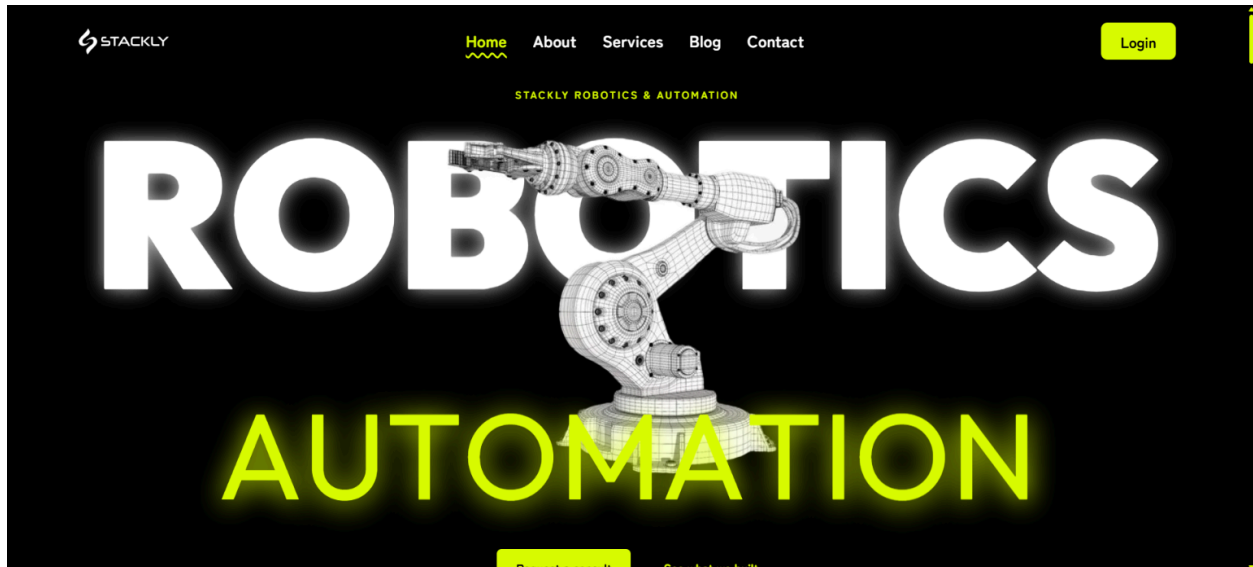
Font Name	CSS Variable	Font Family	Usage Context
Body Font	--font-body	"Outfit", sans-serif	Body text, cards, descriptions, form inputs
Auth Display	--font-display	"Barlow Condensed", "Zen Maru Gothic", sans-serif	Auth page titles, quotes
Auth Body	—	"Inter", "Outfit", sans-serif	Auth page labels, inputs, footers
Font Weights	—	Retro Floral: 900 / Outfit: 400–900 / Zen Maru Gothic: 400, 700 / Barlow Condensed: 400–700 / Inter: 300–700	Clear visual hierarchy across all typographic surfaces

Typographic Scale

Element	Size (clamp)	Weight	Notes
Hero title	32px → 68px	900	Uppercase, tight letter-spacing
Section title orange	30px → 54px	900	Uppercase, primary yellow
Section title (white)	28px → 48px	900	Uppercase, tight letter-spacing
Card title	17px → 22px	700/900	Uppercase
Body text	14px → 17px	400/500	Standard line-height 1.6
Eyebrow / badge	11px → 13px	600/800	Uppercase, wide letter-spacing

Overall Layout & Wireframe

Public-Facing Architecture



Home Page (index.html)

Hero Section — Robotics / Automation (01): Full-screen hero with a giant neon-outlined "ROBOTICS" word behind a robotic arm wireframe image, and a neon-yellow "AUTOMATION" word in front with a text-shadow glow. The two headline words frame a centered robotic arm image in a three-layer composition. A gravity-fall GSAP animation drops the characters into place with a bounce easing. Two CTAs sit at the bottom ("Request a consult" solid yellow, "See what we built" text link).

Machine Section (02): Two-column header with a left title ("MACHINES THAT FINISH THE SHIFT YOU STARTED.") and right description, followed by a full-width framed `machine.webp` image with a yellow border and glow.

Cycle Section (03): Split header ("FROM FLOOR WALK TO FIRST CYCLE" with a yellow-highlighted line), then a large robotic arm deployment diagram with four absolutely-positioned callouts (Assess the Floor, Design the Cell, Build & Integrate, Commission & Handover) connected to the arm by SVG polyline leaders. Each callout carries a yellow title and a description.

Systems Section (04): Three-card grid with a yellow-accent featured card (Articulated Arms) flanked by two gray cards (Autonomous Mobile Robots, Vision & Control). Each card has an icon (custom SVG), a title, a description and a spec list. Cards invert color on hover — the yellow card goes dark with a yellow title, the gray cards go yellow.

Built For The Lines (05): Six-card industry grid with background photography and dark overlay captions (Automotive, Electronics, Warehousing, Food & Beverage, Pharma, Metal Fabrication).

Transform Section (06): Two-column split with an editorial headline on the left and a framed `transform.webp` image on the right.

Number Stats Section (07): Centered header ("Scale that shows up on the floor.") with a four-card grid of animated counters (2,400 work cells, 99.4% uptime, 18 countries, 72-hour parts dispatch). Cards lift and invert to yellow on hover.

Why Stackly Section (08): Two-column split with a facility photograph on the left and an editorial block on the right containing an eyebrow, a title with an italic emphasis, and a three-item icon list (Simulation-first, One control layer, Uptime as the metric).

Case Study Section (09): Two-column split inside a dark rounded card. Left: eyebrow, large pull quote ("The cell has run three shifts a day for six years without a rebuild."), and an author block with an avatar. Right: a facility image.

Final CTA Section (10): Full-width rounded banner with a background photograph, dark gradient scrim, eyebrow, large display title ("Ready to automate *the line?*"), lead text, and a "Request a consult" CTA.

Footer (11): Five-column footer with brand description, social icons, navigation, systems, support, and contact information.

Preloader: A 1.5-second brand preloader overlay with the logo and a spinning yellow ring.

Styles: index.css & global.css

Scripts: Embedded JavaScript within index.html (gravity fall, popcorn pop, preloader, mobile drawer, header scroll state, reveal observer, counters)

Assets: /assets/home/.webp, /assets/remote/.webp, /logo.webp

About Page (/html/about.html)

Hero — About Stackly (01): Multi-layer hero composition with a giant outlined "BUILT TO OPERATE" word behind a centered robotic arm image, and a neon-yellow "WHERE OTHERS CAN'T" word in front with a text-shadow glow. Below the stage sits a two-column row: a yellow card with four overlapping avatar circles and a "10,000+ people already joined" line, plus a right-side paragraph.

How It Works Section (02): Centered header ("DETAILED GUIDE ON HOW THE SYSTEM WORKS AND OPERATES"), followed by a 3-column layout with two cards on the left (DATA GENERATED, DATA PROCESSING), a centered AI head graphic, and two cards on the right (DATA STORED, ACTIONABLE INSIGHTS).

What We Hold Ourselves To Section (03): Split header with a large left title and right subtext, followed by a full-width feature card (Built For The Floor, Not The Demo) and a three-card bottom row (Uptime Over Showmanship, Plain Documentation Always, We Answer Our Own Service Line). Each card has background photography with a dark gradient overlay.

Twelve Years, In Order Timeline (04): Centered header and a vertical timeline with a neon yellow dot spine at the center and a large wireframe sphere graphic in the background. Five milestones (2014, 2016, 2019, 2021, 2026) alternate left and right with year labels, titles and descriptions.

Our Testing Facility Section (05): Two-column split with a left editorial block (eyebrow, title, description, three facility stats: 50k+ Sq. Ft., 100% Pre-tested, <0.01s Latency) and a right framed facility image.

Leadership Section (06): Centered header ("ENGINEERED BY PRACTITIONERS") and a three-card leader grid with portraits, names, roles, and bios.

Final CTA Section (07): Same full-width CTA banner pattern as the homepage.

Styles: about.css & global.css

Scripts: Embedded JavaScript within /html/about.html (gravity fall on hero title, popcorn pop on section headings, mobile drawer, preloader, reveal observer)

Assets: /assets/about/.webp, /assets/remote/.webp, /logo.webp

Services Page (/html/services.html)

Hero — Services (01): Two-column hero with a left content column (title "IDEAS THAT PUSH *INDUSTRIES FORWARD*" with a yellow-highlighted second line, description, subtext, and a "Get a scope" CTA) and a right robotic arm image. A large neon-yellow diagonal wedge sits behind the whole composition with a `clip-path` polygon.

Smarter Industry Solutions Section (02): Full-width dark card with background photography, a yellow pill tag ("Smarter Industry Solutions") and a bottom description overlay. Includes a dark gradient scrim.

Floor Walk to First Cycle Section (03): Split header and a large deployment diagram with four absolutely-positioned callouts connected by SVG leaders.

End-to-End Automation Design Section (04): Two-column header (title + subtext on the left, "Get started" CTA on the right), a full-width `end.webp` banner image with a yellow border, and a three-card grid below (Autonomous System Deployment, Proven Real-World Deployments, Seamless Infrastructure Integration). Cards alternate yellow/gray themes with hover inversion.

Specialized Industrial Cell Solutions Section (05): Centered header and a three-card grid of pre-configured work cells (High-Speed Palletizing, 3D Vision Bin Picking marked POPULAR, Autonomous Mobile Robots). Each card has an icon, title, description, and spec list.

Capabilities & SLA Section (06): Centered header and a vertical roadmap with three highlighted steps (24/7 Remote Telemetry, 72-Hour Parts Dispatch SLA highlighted, On-Site Operator Training). Each step has a number, a badge, a title and a description.

Final CTA Section (07): Full-width CTA banner with a "Request a consult" CTA.

Styles: `services.css` & `global.css`

Scripts: Embedded JavaScript within `/html/services.html` (gravity fall on hero title, popcorn pop on section headings, mobile drawer, preloader, reveal observer)

Assets: `/assets/service/.webp`, `/assets/remote/.webp`, `/logo.webp`

Blog Page (`/html/blog.html`)

Blog Hero Section (01): Two-column hero with a left content column (title "IDEAS THAT PUSH *INDUSTRIES FORWARD*", description, tagline, "Request a quote" CTA) and a right visual column with a yellow `clip-path` polygon background and a `hero.webp` image. The hero title uses gravity fall.

Latest Articles Section (02): Section header with a large yellow title and a set of filter tabs (ALL, Reliability, Mobile robots, Software, ROI & planning). Below sits a six-card articles grid using a 3-column layout with mixed wide/narrow spans. Articles filter in real time when a tab is clicked. Each card has an image, a category eyebrow, an uppercase title, a description and a meta line.

Bracket Story Section (03): Two-column split with a left editorial block (large title "HOW A FORTY-CENT BRACKET COST \$2.1 MILLION", two body paragraphs, "Get

started" CTA) and a right blueprint-style wireframe image.

Featured Case Study Section (04): Two-column split with a left image (`facility.webp`) carrying a "FEATURED CASE STUDY" badge, and a right content block with a subtitle, title ("REDUCING CYCLE TIME BY 38% IN HIGH-SPEED LINES"), a description, a three-metric row with animated counters (38% Cycle Time Saved, 99.9% Cell Up-Time, 4.2 Mo Full Payback) and a "Read full case study" CTA.

Technical Guides & Topics Section (05): Centered header ("TECHNICAL GUIDES & TOPICS") and a four-card grid (Inverse Kinematics, 3D Stereo Vision, ISO 10218, EtherNet/IP). Each card has an icon, a tag, a title, a description and an "Explore Guide" link.

Automation Pulse Newsletter Section (06): A single rounded newsletter card with a yellow left accent, an eyebrow badge, a title, a description, and a form (email input + Subscribe button) that validates the email, clears the field, and routes to 404.html.

CTA Banner Section (07): Full-width CTA banner with a background photograph, dark overlay, and a display title ("TAILORED TECHNOLOGIES THAT EMPOWER GLOBAL INDUSTRIES FORWARD"), description, and "Get started" CTA.

Styles: `blog.css` & `global.css`

Scripts: Embedded JavaScript within `/html/blog.html` (article filtering, gravity fall on hero title, popcorn pop on section headings, case-study counters, newsletter validation, mobile drawer, preloader, reveal observer)

Assets: `/assets/blog/*.webp`, `/assets/about/facility.webp`, `/logo.webp`

Contact Page (`/html/contact.html`)

Contact Hero Section (01): A centered hero with two typography lines ("TELL US YOUR VOLUMES" in white and "WE'LL TELL YOU WHAT IT TAKES" in yellow), followed by a 5-ring robotic arms stage. The stage has a centered large ring (540×540) flanked by left and right columns each with two smaller rings — all floating on subtle staggered animations. Below the stage sits a sub-bar with a paragraph and a "Contact us" CTA.

Contact Form Section (02): Two-column grid with a form card on the left (name, email, phone, business type, company, location, message, privacy checkbox) and

a direct contact card on the right (email, phone, address — each as a link to 404.html). The form has custom validation with inline error messages.

FAQ Section (03): Six-item accordion with click-to-expand cards, animated plus-to-close icons, and a highlighted active state.

Engagement Process Section (04): Four-step process grid (20-Min Discovery Call, 3D Digital Twin Simulation, Build & FAT Validation, Turnkey On-Site Launch). Each card has a large numbered badge, an icon, a title and a description.

Map Section (05): Full-width Google Maps iframe embed with a border and hover-lift filter.

CTA Banner Section (06): Rounded CTA banner with a display title, description, and two CTAs ("Schedule a 20-Min Call" and "Explore Services").

Styles: contact.css & global.css

Scripts: Embedded JavaScript within /html/contact.html (contact form validation, FAQ accordion, gravity fall on hero title, popcorn pop on section headings, mobile drawer, preloader, reveal observer)

Assets: /assets/contact/*.webp, /logo.webp

Authentication Portals (/html/login.html & /html/signup.html)

Split-Screen Design: Both pages use a two-column split layout:

- **Login:** Photo panel (with a background hero photograph, dark scrim, brand mark top-left, and a bottom-left italic quote) on the left; form panel on the right with a "Back to Home" chip top-right
- **Signup:** Form panel on the left; photo panel on the right

Login Form Controls: Email, password with reveal toggle, and a custom role-select dropdown (Operator / Engineer, Admin / Integrator) built from a button trigger plus a positioned listbox. Full keyboard navigation (Arrow Up/Down, Enter, Space, Escape). A "Forgot password?" auxiliary link completes the form.

Signup Form Controls: Full name, email, password, confirm password (both with individual reveal toggles), and a custom role-select dropdown.

Client-Side Validation: Both forms validate required inputs, email regex patterns, password length (min 6), password confirmation matching, and role selection. Errors surface inline beneath each field with a red `is-invalid` border state.

Role-Based Routing: On successful login, the email and role are stored in `localStorage` under `stackly_user_email`, `stackly_operator_email`, `stackly_user_role`, `stackly_operator_role`, `stackly_user_name` and `stackly_operator_name`, then the browser redirects to `admin.html` (for Administrator) or `user.html` (for Operator). Signup does not persist — it clears the form and routes to login.

Pageshow Handling: Both forms register a `pageshow` listener that clears all fields, resets dropdown selections, and removes error states whenever the browser restores the page from `bfcache`.

Social Login Placeholders: Google and Apple social buttons route to `404.html`

Styles: `/css/login.css` & `/css/signup.css`

Scripts: Embedded JavaScript within `/html/login.html` & `/html/signup.html`

Assets: `/assets/login_page.webp`, `/assets/signup_page.webp`, `/logo.webp`

Error Page (404.html)

Self-Contained Error Page: A full-viewport single-screen error page with a horizontal scanline grid backdrop, a radial signal glow at the top-right, and a two-column layout:

- Left: a massive `404` numeral in Georgia serif italic with a yellow glow at 35% opacity
- Right: an eyebrow ("Error / Signal Lost"), a display title ("This route has been *decommissioned*."), a supporting paragraph, two actions ("Home" yellow pill, "Go Back" outlined pill), and a quick links row

No Scroll: The 404 page is built to fit exactly in the viewport on all screen sizes, with `overflow: hidden` on the `html` and `body`.

Directional Arrow Motion: Both buttons use opposing hover behaviour — the arrow in "Go Back" slides left, and "Home" doesn't have an arrow.

Reduced Motion: All entrance animations are skipped when `prefers-reduced-motion: reduce` is set.

Styles: Inline `<style>` block within `404.html` (no separate CSS file)

Scripts: Minimal "Go Back" handler with fallback to `index.html` + GSAP entrance timeline

Assets: `/logo.webp`

Internal Architecture (Dashboards)

Both dashboards operate on a Single Page Application (SPA) architecture without page reloads. Using dynamic JavaScript event listeners, the implementation toggles panel visibility on the fly, updates the view-section display properties, and re-runs chart animations on the newly activated panel to deliver a fast, seamless tab-switching experience. Both dashboards adopt a **fixed sidebar + independently scrolling main** layout, with a full-screen slide-in mobile drawer replacing the sidebar on viewports $\leq 768\text{px}$.

Admin Console (/html/admin.html) — Fleet Command

Fixed Sidebar (Desktop): Deep surface sidebar with brand mark, an "Administrator" chip, and grouped navigation:

- **Operations:** Control Room (default), Fleet, Telemetry
- **Team & Autonomy:** Operators, Autonomy
- **System:** System
- Each item carries a symbol icon, a label, and optional badge (e.g. 248)
- Sidebar foot holds a signed-in-as card and a logout button

Fixed Mobile Header: Small mobile header (visible $\leq 768\text{px}$) with a hamburger button, brand mark, spacer and avatar initial

Full-Screen Mobile Drawer: Sidebar slides in from the left with a scrim overlay. Includes body scroll locking and Escape key dismissal. Double-click on the brand collapses the sidebar on desktop.

Dynamic Tab Views: Organized into six separate content panels:

Control Room:

- Four KPI cards (Tasks Completed, Task Success Rate, Active Robots, Avg Cycle Time)
- Line chart (Fleet Throughput, last 7 days) drawn with SVG path with a `drawChart` keyframe animation
- Donut chart (Fleet by Robot Type: Articulated Arms 70%, Mobile Robots 20%, Vision Cells 10%)

- Recent Deployments table with three entries
- Recent Activity feed with three entries
- Quick actions cards

Fleet:

- Four KPI cards (Total Deployed Units 248, Operational 223, Needs Maintenance 07, Avg Fleet Health 94.8%)
- Six-row Unit Directory table with unit ID, facility, class/firmware, total cycles, health bar, and status badge

Telemetry:

- Four KPI cards (Task Success Rate, Mean Time Between Faults, Avg Cycle Time, Fleet OEE)
- Weekly Throughput bar chart with animated bars
- Diagnostic Breakdown donut chart
- Unit Class Performance table with three rows (Articulated Arms, Mobile Robots, Vision Cells)

Operators:

- Four KPI cards (Active Operators, Incidents/Operator, Avg Resolution Time, Team Rating)
- Three operator cards (Sarah Kim, James Davis, Emma Miller) with avatars, roles, status badges, and productivity progress bars
- Operator Scorecard table with three rows

Autonomy:

- Four KPI cards (Auto-Resolved Tasks, Autonomy Rate, Operator Hours Saved, Decision Accuracy)
- Three automation cards (Auto-Recovery, Smart Dispatch, AI Diagnostics) with toggles
- Autonomy Adoption bar chart (6 weeks)

System:

- Two settings cards (Facility & Telemetry Configuration, Security & Access Control)
- Integrations & API Gateway card

Identity Resolution: Reads `stackly_operator_email` from `localStorage` and displays the local-part as a first name. Falls back to `admin@stackly.example`.

Logout: Clears all storage keys and routes to `login.html`.

Styles: Inline `<style>` block within `/html/admin.html`

Scripts: Embedded JavaScript within `/html/admin.html`

Assets: `/logo.webp`

Operator Console (`/html/user.html`)

Fixed Sidebar (Desktop): Deep surface sidebar with brand mark and grouped navigation:

- **Operations:** Control Room (default), Work Orders (badge 4), Field Manual
- **Account:** Operator Profile, Preferences
- Sidebar foot holds a mini user profile card and a logout button

Fixed Mobile Header: Same mobile header pattern as admin

Full-Screen Mobile Drawer: Same slide-in pattern as admin

Dynamic Tab Views: Organized into six separate content panels:

Control Room:

- Four KPI cards (Active Units 04, Completed Tasks 28, Avg Cycle Time 42s, Fleet Uptime 99.2%)
- Chart.js-powered line chart (Site activity, last 7 days)
- Chart.js-powered donut chart (Work order status)
- Recent work orders table with three entries
- Three quick-action cards (Request service, Browse field manual, Talk to engineering)

Work Orders:

- Four KPI cards (Open Orders 04, In Progress 02, Awaiting Parts 01, Resolved 30 Days 24)
- Search toolbar and "New work order" CTA
- Six-row Work Orders table with ticket avatar, title, subsystem, priority badge, dates, and status

Support:

- Two-column layout with a channel list on the left (Dispatch, Engineering, Parts & Warranty) and a chat-style conversation pane on the right
- The message form appends a user message and simulates an agent reply after 900ms

Field Manual:

- Search bar with popular tag chips (ERR-502 Gripper, LiDAR Calibration, Modbus TCP, EtherCAT Bus, ISO 10218 E-Stop)
- Six knowledge cards (Commissioning & Setup, Fault Codes & Diagnostics, Safety & Compliance, PLC & Fieldbus Integration, Vision & AI Navigation, Spare Parts & Maintenance), each with an icon, title, description and article count
- A Frequent Fault Codes table (ERR-502, ERR-314, ERR-109) with action steps

Operator Profile:

- A profile card with avatar, name, email, role, and clearance details
- A certifications card (6-DOF Arms, AMR Fleet, ISO 10218 E-Stop)
- A form card for personal information (First name, Last name, Email, Phone, Facility, Specialization)
- A security card (Current Security Key, 2FA Authentication)

Preferences:

- Two settings cards (Notifications & Alerts, Console Display & Audio)
- A Telemetry Thresholds & Shift Reporting card with three more toggles

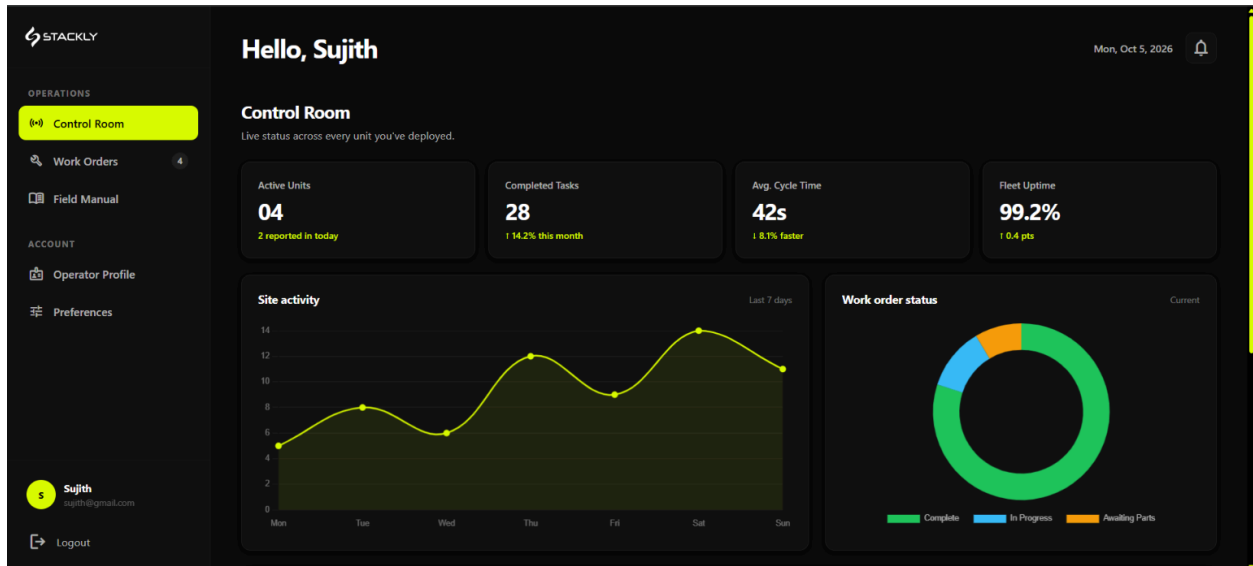
Identity Resolution: Reads `stackly_operator_email` and `stackly_operator_name` from `localStorage`. Role-based guard redirects admin users to `admin.html`.

Logout: Clears all storage keys and routes to login.html.

Styles: Inline `<style>` block within /html/user.html

Scripts: Embedded JavaScript within /html/user.html

Assets: /logo.webp



Animation & Performance Standards

Gravity Fall Hero Text Animation: The platform's signature hero animation splits a heading into individual character spans and drops them in from above with `y: -300, opacity: 0, stagger: 0.04, duration: 0.8, ease: 'bounce.out'`. This produces a distinctive mechanical typing effect — each letter falls into place and rebounds — reinforcing the industrial, robotic tone. The splitter operates on `.text-robotics`, `.text-automation`, `.text-built-operate`, `.text-where-others`, `.blog-hero-title .title-line-1/2`, `.service-hero-title .title-line-1/2`, and `.contact-hero-title .title-line-1/2`.

Popcorn Pop Section Heading Animation: Section headings are split into character spans and animated in with `scale: 0, y: 30, rotation: random(-20, 20), stagger: { each: 0.04, from: 'random' }, duration: 0.4, ease: 'back.out(2)'`. This creates a "popcorn" effect that reinforces the operational, energy-driven tone of the brand.

Directional Reveal Motion System: Content blocks across the site use the `[data-anim]` attribute system (`left`, `right`, `up`, `down`, `fade`) with a shared `cubic-bezier(0.22, 1, 0.36, 1)` easing curve over 0.9s. Elements are activated by an `IntersectionObserver` with a `data-delay` attribute (in seconds) applied per-element.

Lenis Smooth Scroll: A shared `lenis-init.js` file loads the Lenis library from CDN and initialises a lerped `requestAnimationFrame` loop with `duration: 1.2` and `easing: (t) => Math.min(1, 1.001 - Math.pow(2, -10 * t))`. Lenis replaces the browser's native scroll physics across every public page.

Hero Rotation & Marquee Loops: The ticker section on the homepage and the testimonial marquee use duplicated flex groups animated with a linear `translateX(-50%)` keyframe. Both pause on hover. The about page's team marquee uses the same technique.

Interactive ROI / Case Study Counters: The blog page's featured case study includes a three-metric counter row that animates from 0 to its target when scrolled into view using `requestAnimationFrame` and a `1 - Math.pow(1 - p, 3)` easing curve.

Dashboard Chart Animations: Both dashboards build their charts from static markup and CSS/JS:

- Line charts in admin are SVG paths with a `drawChart` keyframe animating `stroke-dashoffset`
- Bar charts use `transform: scaleY()` over 0.9s with staggered delays
- Donut charts use `conic-gradient` backgrounds
- KPI counters use GSAP + ScrollTrigger with a `power2.out` easing
- Operator console line + donut charts are rendered using the Chart.js library

Preloader: All public-facing pages display a 1.5-second brand preloader with a spinning yellow ring before the content is revealed.

Scroll Lock: The mobile drawer applies `body.menu-open` and `html.no-scroll` states to freeze background scrolling.

Reduced-Motion Compliance: Every stylesheet includes a `prefers-reduced-motion: reduce` block that disables transforms, transitions, GSAP animations, and marquee animations.

Authentication & Error Handling

Client-Side Security & Validation

Form Validation Rules: Validates input requirements across authentication fields, checking email regex patterns (`/^[^\s@]+@[^\s@]+\.[^\s@]+$/`), enforcing minimum password lengths (6 characters), verifying role selections, ensuring password confirmation matching on signup, and requiring the privacy policy acknowledgement on the contact form.

Real-time Error Guidance: Prevents default submission on invalid entries, displaying targeted error messages beneath each field, changing the field border colour to a red error state, and moving focus to the first invalid field.

Custom Dropdown Components: Both login and signup use a custom dropdown built from a `<button>` trigger and a positioned `<ul role="listbox">` containing `<li role="option">` items. Full keyboard navigation (Arrow Down/Up for focus movement, Enter/Space to select, Escape to close, Tab to close and move on) is supported, and `aria-selected` state is synchronised with a hidden input value on selection.

Password Reveal Toggles: Both password fields (and both signup confirmation fields) have individual reveal toggles that switch the input `type` between `password` and `text`, and update the button's `aria-pressed` state and icon.

WebKit/MS Native UI Removal: Edge, Chrome, and Safari native reveal buttons are hidden with `::-ms-reveal`, `::-ms-clear`, `::-webkit-credentials-auto-fill-button`, and `::-webkit-strong-password-auto-fill-button` selectors.

Autofill Background Suppression: Custom autofill styles apply a solid dark background using `-webkit-box-shadow: 0 0 0 1000px var(--field-bg) inset !important` so the browser's yellow autofill background does not clash with the dark theme.

Local Session Storage: On successful login, the validated email, derived local-part name, and selected role are stored in `localStorage` under `stackly_user_email`, `stackly_operator_email`, `stackly_user_name`, `stackly_operator_name`, `stackly_user_role`, and `stackly_operator_role`. Both dashboards read these keys to populate the sidebar identity card and header avatar, with hard fallbacks if storage is unavailable.

Role-Based Portal Routing: Login redirects to `admin.html` for the Administrator role and `user.html` for the Operator role. Signup does not persist — it clears the form and routes to `login.html`.

Form Submission Prevention: All forms (login, signup, contact, newsletter) prevent default submission with `event.preventDefault()` and handle validation

programmatically before any redirect.

Pageshow Reset: Both authentication pages register a `pageshow` listener that resets all fields, clears all errors, and resets dropdown selections whenever the browser restores the page from bfcache.

Password Confirmation: On signup, validates that both password fields match before allowing submission, with clear error messaging for mismatched passwords.

Responsive & Performance Methodologies

Fluid Layouts & Adaptive Breakpoints: Employs flexible CSS grid structures, flexbox containers, and standard media queries across all viewports (1200px, 1100px, 1024px, 992px, 900px, 820px, 768px, 680px, 640px, 600px, 560px, 550px, 480px, 380px, 300px breakpoints) to ensure seamless UI adaptation from mobile phones up to high-resolution desktop displays. The hero collapses its multi-layer composition into a single-column stack on mobile, the service grid collapses from a 3-column grid to a single-column layout, the pricing table stacks vertically, and the dashboards transform their fixed sidebars into full-screen slide-in drawers.

Full-Screen Mobile Drawer Pattern: Both dashboards and the public site use a full-screen slide-in drawer on mobile. Each drawer includes body scroll locking, Escape key dismissal, and automatic close when a navigation item is selected.

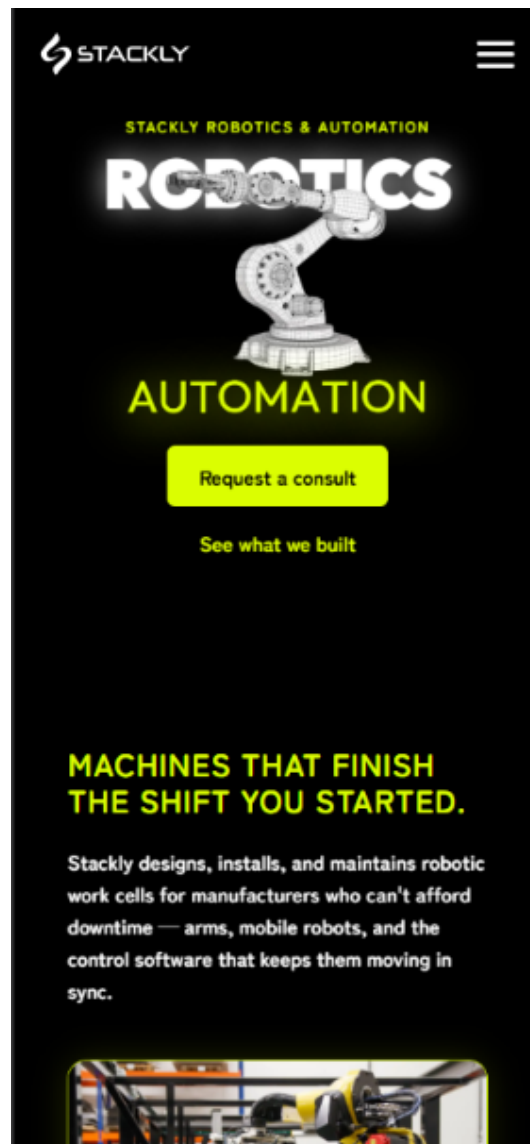
Hardware-Accelerated Motion: Uses `transform` and `opacity` animations throughout, with a shared easing curve. Dashboard chart animations use CSS transitions on transform properties and SVG attributes.

Optimized Asset Delivery: Uses modern WebP image formats across all hero banners, warehouse imagery, robotic arm photography, team portraits and facility photography, combined with explicit `loading="lazy"` attributes on non-critical imagery.

Dashboard Performance: Both dashboards build their charts from static markup rather than generating them at runtime. Chart animations are triggered by an `IntersectionObserver` and skipped when `prefers-reduced-motion: reduce` is set.

Custom Thin Scrollbar: A 7px custom scrollbar with a yellow thumb and black track matches the platform's industrial aesthetic across all pages.

Reduced-Motion Compliance: Every stylesheet includes a `prefers-reduced-motion: reduce` media block that disables transforms, transitions, and marquee animations.



SEO & Asset Optimization

Deferred & Asynchronous Resource Loading: All Font Awesome, Material Symbols, Google Fonts, GSAP, ScrollTrigger, Chart.js and Lenis resources are loaded with `preconnect` hints for the font hosts. GSAP and Chart.js are loaded at the end of the body to avoid blocking the initial render. Non-critical images are marked with `loading="lazy"`.

Semantic HTML Markup & Document Metadata: Structured using standard structural tags (`<header>` , `<nav>` , `<main>` , `<section>` , `<aside>` , `<footer>`) with explicit `<title>` tags (e.g., "Stackly - Robotics & Automation", "About Us - Stackly Robotics & Automation", "Blog - Stackly Robotics & Automation") and responsive viewport meta elements on every page. Dashboards additionally set `<meta name="robots" content="noindex">` .

Accessibility Metadata: Every page includes ARIA roles for tab panels (`role="tablist"` , `role="tab"` , `role="tabpanel"`), `aria-selected` state, `aria-expanded` state on all toggles, `aria-controls` on all interactive elements, `role="listbox"` and `role="option"` on custom dropdowns, `role="dialog"` and `aria-modal="true"` on the mobile drawer, and detailed `aria-label` text on icon-only buttons and charts.

Clean URL Structure & Navigation: Implements intuitive, SEO-friendly URL patterns across all pages with consistent internal linking through the global navigation header and footer.

Asset Formats: Leverages optimized WebP media files across all hero images, warehouse photography, robotic imagery and team portraits, significantly minimizing file weights compared to traditional PNG or JPEG formats.

Future Scalability Insights

While the current frontend architecture delivers a responsive, visually compelling user experience and meets immediate robotics platform requirements, enterprise scaling requires full backend integration and enhanced data persistence.

Backend Framework Integration

Transitioning the static HTML, JavaScript, and tab-based architecture into a framework like React, Angular, or Vue.js — paired with a Node.js, Python, or .NET Core API backend — will enable robust state management, modular component reusability, and efficient routing across complex dashboard layouts.

Database & Secure Storage

Migrating temporary localStorage session data (user profiles, role-based permissions, login status, work order records, telemetry logs) into a relational database such as PostgreSQL or MySQL will securely store operator profiles, fleet

records, work order histories and telemetry streams, ensuring data persistence and enabling advanced query capabilities.

Real-Time Telemetry Integration

Building automated workflows between the operator console and live robot telemetry (motor current, joint temperatures, cycle timings) will enable true real-time visibility of every deployed unit's kinematic health without requiring page refreshes.

Fleet Management Integration

Integrating the admin console with a fleet manager (VDA 5050, ROS2, or vendor-specific APIs) will enable automated dispatch, route assignment, and firmware deployment directly from within the platform.

Work Order Automation

Building a full work order lifecycle — from operator submission through dispatch, technician assignment, parts ordering, and closure — will replace the current demonstration workflow with a production-grade ticketing system.

Spare Parts Integration

Connecting the field manual and work order console to a spare parts inventory and dispatch system will enable automated parts lookup, availability checking, and 72-hour SLA tracking.

Predictive Maintenance Integration

Extending the autonomy panel to ingest live motor current and vibration data will enable predictive fault detection and scheduled maintenance planning without requiring engineer intervention.

Multi-Region & Multi-Language Support

The architecture can be extended to support multiple facilities and working languages (English, German, Japanese), enabling Stackly to manage robot fleets across geographies from a unified platform.

Audit Trail & Compliance Reporting

Building a full audit log with filterable event types, actor attribution, and export capabilities will support ISO 10218, ISO 9283 and CE compliance reporting requirements.

Field Manual CMS

Extending the platform to publish field guides and fault code references through a structured CMS will enable engineering teams to update documentation without manual HTML edits.

Mobile Field App

Building a lightweight mobile app for field engineers will enable on-the-ground work order handling, parts scanning, and fault code lookup that syncs back to the console in real time.

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